

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

YARN MANUFACTURE-II

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	State what is Flyer Leading	M 1.01	U
2	Mention the function of spacer	M1.03	U
3	List the function of separator in Ring Frame	M1.03	U
4	State the relation between Rotor speed and Rotor Dia	M2.02	U
5	List any two types of Rotor Grooves	M2.02	U
6	Differentiate between dry doubling and wet doubling	M3 01	U
7	State what is a balanced yarn	M3 02	U
8	Express the relation between TPI and TM	M4 01	U
9	Find out the traveller speed in meter per second if the spindle speed is 18000 rpm and ring diameter is 40 mm	M4 02	A

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Write the functions of spindle, flyer and presser	M 1.01	U
2	Briefly state how traveller is numbered	M1 03	U
3	List the advantage of compact yarn over the ring yarn	M2 01	U
4	With a schematic diagram explain the principle of Airjet spinning	M2 04	U
5	Differentiate Ply, Cord and Cabled Yarn	M3 01	U
6	Write the Characteristics of Slub Yarn and Loop Yarn	M3 03	U
7	Find out the maximum spindle speed practical of a ring frame using 36 mm dia ring with maximum practical traveller speed is 45m/s.	M4 02	A

8	Find out the resultant count of a 3 fold yarn consisting of 24 ^s , 30 ^s & 36 ^s .	M4 04	A
9	Explain the objects of building mechanism of the ring frame	M1 03	U
10	Mention the objects of reeling and state the coun of the yarn if the number of knots in a bundle is 20.	M3 04	A

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain the working of Speed Frame with a neat sketch	M1 01	U
OR			
2	Explain the need for variable speed spinning in Ring Frame and explain the VPS system	M1 03	U
3	Illustrate the working of elite compact spinning with a sketch	M2 01	U
OR			
4	Describe the yarn formation in DREF-II friction spinner with a sketch	M2 03	U
5	Compare C Traveller and Elliptical Traveller and explain Running in period of ring	M1 03	U
OR			
6	Explain the SIRO spinning with neat sketch	M2 01	U
7	Describe the working of TFO with neat sketch	M3 03	U
OR			
8	List the objectives of doubling and with sketches explain English and Scotch system of wet doubling	M3 01	U
9	Find out the production per simplex per 8 hour in Kgs from the given details; Spindle speed- 1300 rpm, TM-1.2, Hank of Sliver- 0.16, Draft-10, Efficiency-90%, Number of Spindles-108	M4 01	A
OR			
10	Determine the production per ring frme per 8 hour in kgs from the details given; Spindle speed-21000 rpm, TM-4.6, Hank of Roving-2.4, main Draft Constant-7000, main Draft Wheel-28T, Break Draft-1.2, Efficiency-92%, Number of Spindles/frame- 1008	M4 03	A
11	With a neat sketch explain the working of a fancy doubler	M3 03	U

OR			
12	Find out the production/8hour of a ring doubler from the details given; Counts doubled-20 ^s , 24 ^s Number of Spindle-400, Efficiency-90%, Delivery Roller Dia-2", Delivery Roller Speed-160 rpm	M4 04	A